III. Conditional Probability and Independence (GSE 518)

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1. Conditional Probabilities

Idea: Sometimes we can *update* the probability of an event based on *new information.*

Definition:

If $A \subseteq S$ and $B \subseteq S$ and $P(B) > 0$, then the conditional probability of A given B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example:

Flip a coin 2 times:

$$S = \{HH, HT, TH, TT\}$$

Let $A =$ getting both heads, and $B =$ at least one head:

$$A = \{HH\}, \quad B = \{HH, HT, TH\}$$

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{3}{4}$$

*MSQE Alumni Hunter Happersett contributed significantly to writing these lecture notes in Latex.

What is $P(A|B)$? (Note that since $A \subseteq B, A \cap B = A$)

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Notes:

i). We can think of $P(A|B)$ as updating the sample space from S to B.

$$S' = B = \{HH, HT, TH\} \Rightarrow P(A|B) = \frac{1}{3}$$

ii). **If A and B are disjoint**, $A \cap B = \emptyset$, then:

$$P(A|B) = P(B|A) = 0$$

iii). **For any B with $P(B) > 0$, $P(A|B)$ is a probability function** thus all Kolmogorov's axioms must hold true:

$$0 \leq P(A|B) \leq 1, \quad P(A|B) + P(A^c|B) = 1, \quad \text{etc.}$$

Note: Be careful as $P(A|B)$ and $P(A|B^c)$ are not necessarily complementary, i.e., in general $P(A|B) + P(A|B^c) \neq 1$

2. Some Uses of Conditional Probability

1) To calculate probability of joint events

If $P\left(\bigcap_{i=1}^{n} A_i\right) > 0$:

$$P\left(\bigcap_{i=1}^{n} A_i\right) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \cdots P\left(A_n|\bigcap_{i=1}^{n-1} A_i\right)$$

Special case:

$$P(A \cap B) = P(B) \cdot P(A|B)$$

Example:

Suppose you have a box with r red balls and b blue balls. You sequentially pick 3 at random without replacement. What is the P(all are red)? (Assume $r \geq 3$)

Let r_i be the i-th ball drawn is red:

$$P(\text{all red}) = P(r_1 \cap r_2 \cap r_3)$$

$$= P(r_1) \cdot P(r_2|r_1) \cdot P(r_3|r_1 \cap r_2)$$

$$= \left(\frac{r}{r+b}\right) \cdot \left(\frac{r-1}{(r-1)+b}\right) \cdot \left(\frac{r-2}{(r-2)+b}\right)$$

2) Law of Total Probability

Let $A_1, A_2, \ldots, A_n$ be a partition of S. Then for any event $B \subseteq S$:

$$P(B) = \sum_{i=1}^{n} P(B|A_i) \cdot P(A_i)$$

(Result of Kolmogorov's axiom 6) *partition*

Example:

$A_1 = \text{males}, \quad A_2 = \text{females}, \quad B = \text{white}$

$$P(\text{white}) = P(\text{white}|\text{females}) \cdot P(\text{females}) + P(\text{white}|\text{males}) \cdot P(\text{males})$$

3) Bayes' Theorem/Rule

Let $A_1, A_2, \ldots, A_n$ be a partition of S. Then for any event $B \subseteq S$:

$$P(A_i|B) = \frac{P(B|A_i) \cdot P(A_i)}{\sum_{i=1}^{n} P(B|A_i) \cdot P(A_i)}$$

This is an "inversion theorem". We have $P(B|A_i)$ and want $P(A_i|B)$.

This is equivalent to our other formula for Bayes' Theorem but now in the context of partitions (note that the denominator is just the Law of Total Probability).

Rewriting the formula above:

$$\frac{\frac{P(B \cap A_i)}{P(A_i)} \cdot P(A_i)}{P(B)} = \frac{P(B \cap A_i)}{P(B)}$$

Bayes' Theorem Example: Performance of Diagnostic Test

Let $D =$ "disease" and $\neg D =$ "not disease". A test result is either $+$ or $-$.

Manufacturer gives "sensitivity" of test:

$$P(+|D) = 0.9$$

and "specificity" of test:

$$P(-|\neg D) = 0.95$$

Finally, suppose $P(D) = 0.01$ in population.

Question: What is $P(D|+)$?

$$P(D|+) = \frac{P(D \cap +)}{P(+)}$$

but we don't have this...

... so we can apply Bayes' Rule to get it!

Our partition consists of D or $\neg D$ in this case:

$$P(D|+) = \frac{P(+|D) \cdot P(D)}{P(+|D) \cdot P(D) + P(+|\neg D) \cdot P(\neg D)}$$

$$= \frac{(0.9)(0.01)}{(0.9)(0.01) + P(+|\neg D) \cdot P(\neg D)}$$

If $P(D) = 0.01$ in population, $\Rightarrow P(\neg D) = 0.99$

and

$$P(+|\neg D) = P(+|\neg D) + P(-|\neg D) = 1$$

Careful, it is not $P(+|\neg D) + P(+|D) = 1$

$$\Rightarrow P(+|\neg D) = 1 - P(-|\neg D) = 0.05$$

Thus,

$$= \frac{(0.9)(0.01)}{(0.9)(0.01) + (0.05)(0.99)} = P(D|+)$$

Note:

If we don't have a partition:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \quad \text{(Bayes)}$$

3. Independence

Idea: The occurrence of A does not affect B.

Definition:

Two events are independent if

$$P(A \cap B) = P(A) \cdot P(B)$$

Notes:

- If $P(B) > 0$, then A and B are independent if

$$P(A|B) = P(A)$$

- If A and B are independent, then (A^c, B), (A, B^c), and (A^c, B^c) are independent.

Extension to n events:

A collection of events $A_1, A_2, \ldots, A_n$ are **mutually independent** if for any subcollection $A_{i_1}, A_{i_2}, \ldots, A_{i_k}$ for $k \leq n$:

$$P\left(\bigcap_{j=1}^{k} A_{i_j}\right) = \prod_{j=1}^{k} P(A_{i_j})$$

Example:

For 3 events E_1, E_2, E_3, for independence we would need:

- E_1, E_2
- E_1, E_3
- E_2, E_3

To all be independent:

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$$

$$P(E_1 \cap E_3) = P(E_1) \cdot P(E_3)$$

$$P(E_2 \cap E_3) = P(E_2) \cdot P(E_3)$$

and

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2) \cdot P(E_3)$$

So, the probability of the joint event being equal to the product of the probabilities of the events must hold for every subset.

Careful:

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2) \cdot P(E_3) \not\Rightarrow \text{independent}$$